Personal Vocal Physiology and Acoustics

Moonshot implementation specification - revision 6, September 10, 2026.

1. Research objective and change of direction

Build a system that attempts to infer a person's internal vocal anatomy, dynamic articulation, and eventually vocal-fold mechanics from ordinary audiovisual observations plus actively selected vocal experiments. The resulting personal physical model must generate sound, explain multiple observations with shared anatomy, and predict the consequences of interventions. Singing coaching is the downstream application of that model.

This plan supersedes the current remote repository architecture and scope; existing code is optional reusable material. Revision 6 adds embodied cue-based teaching, dynamic movement mapping and a separate model of learned motor control to the parallel-agent design. It retains the integrated iPhone depth work and physiological reconstruction objective.

This revision supersedes the earlier recommendation to make empirical cue personalization the main product. Failure to recover physiology is an acceptable research outcome. A cue recommendation engine is not a substitute deliverable. Scientific uncertainty determines what we test and how we interpret the result; it does not remove the moonshot from scope.

The core hypothesis is: **a physically constrained model, personalized across multiple tasks and updated through informative interventions, can recover physiologically meaningful individual parameters more accurately than passive observation or a generic anatomical prior.** The full ambition remains internal geometry and tissue mechanics. The first experiment estimates a tractable subset with an extensible physical representation, rather than asserting every internal property is identifiable immediately.

The statement that complete reconstruction is not established belongs in the research motivation. It is neither a veto nor proof of novelty. Prior work jointly estimates anatomy and articulation using VocalTractLab. In that work, validation against directly measured anatomical parameters remained future work. Our proposed contribution is active, multimodal, individual reconstruction with independent tests of anatomy and intervention prediction.[1]

Earlier plan	Revised plan
Primary output: which cue helps this singer	Primary output: executable personal physiological acoustic model with uncertainty
Simulator illustrates a possible explanation	Simulator generates predictions and participates in parameter inference
Camera optional to the central experiment	Camera is an experimental information source, tested against an audio-only condition
Anatomy deferred outside the hackathon	Anatomy inference is the hackathon's critical path
Success means better practice performance	Success means evidence of parameter recovery and prospective physical prediction
Coaching interface gets most engineering time	Forward physics, inverse fitting and experiment design get most engineering time

1. Sun, Huang and Wu. Unsupervised Acoustic-to-Articulatory Inversion with Variable Vocal Tract Anatomy. Interspeech 2022; measured-anatomy validation i...

Working assumptions: two human leads, each coordinating parallel Astra agents, with local development hardware and the available iPhones. Work is ordered by dependencies and acceptance evidence; no hackathon duration or delivery estimates are assumed.

2. What counts as progress, success and failure

Separate four levels of evidence. Publish all completed levels, including negative results, without relabeling a lower level as a higher one.

Level	Experiment	What it establishes
A: recovery in simulation	Hide known parameters, generate diverse observations, infer them, and score recovery on held-out synthetic people.	Implementation correctness and identifiability within that simulator.
B: human prediction	Fit calibration trials, freeze a prediction, then acquire an unseen intervention recording.	Out-of-sample acoustic prediction for that person; not anatomical truth.
C: independent physiology	Compare hidden-state predictions with withheld, modality-appropriate measurements from that person.	Evidence for the specific anatomical or physiological variables measured.
D: coaching benefit	Use the inferred model to choose interventions and test retained learning on new material.	Downstream educational value; not a replacement for A-C.

The minimum meaningful hackathon build runs A and attempts B with genuine physiology parameters, a working forward simulator, multiple candidate solutions, and a reproducible log. Attempt C during the hackathon only if suitable paired data and permission are already available. Otherwise report C as untested. A failed live fit, non-identifiable parameters, or no gain from active experiments is a valid outcome; a polished coaching demo does not repair that result.

Pre-register two primary comparisons before human evaluation: personalized shared anatomy versus fixed generic anatomy with equally flexible articulation; and active experiment selection versus a fixed protocol under the same recording and computation budgets. Also test audio-only versus audiovisual inference. A population acoustic predictor is a useful additional baseline. Do not let the personalized model win simply by giving it more per-recording free parameters.

Novelty is a research hypothesis, not a claim that no related system exists. A 2026 study predicts MRI-derived contours from a single speaker's speech; that differs from inferring unseen singers' full 3D geometry and mechanics from consumer capture. It is a useful benchmark direction, not proof of the proposed capability.[2]

3. Physical model and parameter hierarchy

Represent the person as a generative physical system, not an unconstrained voice embedding. The model should expose geometry, articulatory state, excitation and capture effects separately. All stored parameters have names, units, bounds, provenance, and an interpretation tied to the equations or simulator.

2. Azzouz et al. Acoustic-to-articulatory inversion preprint, March 30, 2026. Full text and dataset limitations.

```
Shared anatomy theta + tissue parameters phi
                 |
Task / intervention u -> dynamic articulation z(t)
                 |              |
                 +--> coupled source and tract solver --> sound
                                |                          |
                         visible surface model       room / microphone
                                |                          |
                           predicted video            predicted audio

Observed video + audio -> inverse solver -> candidate personal models
                                      -> uncertainty / model mismatch
                                      -> next discriminating experiment
```

Shared anatomy, theta. Parameterized oral/pharyngeal shape, palate/jaw/tongue morphology, tract lengths, lip dimensions, and nasal geometry where supported. These variables are shared across recordings of a person. They do not change merely because the vowel, microphone, or instruction changes. Soft-tissue pose belongs in dynamic articulation; long-term anatomical changes are outside the first study.

Dynamic articulation, z(t). Jaw position, lip aperture/protrusion, tongue configuration, larynx position and velopharyngeal aperture. These vary through a phrase under feasible motion constraints. A requested action is not an exact observation of z(t): the singer can misunderstand or compensate. Visible behavior constrains only what the camera actually observes.

Tissue mechanics, phi, and activation. The target includes vocal-fold geometry, effective mass, stiffness and damping, with activation and driving pressure distinguished from passive properties. A kinematic glottal waveform is an initial excitation model, not a measurement of these quantities. Add a supported self-oscillating model as an explicit mechanics experiment; do not rename pulse-shape controls as tissue stiffness.

Capture and nuisance variables, eta. Microphone response, gain, room, pose, timing offsets and session state. Constrain these by calibration and repeated capture. Do not let an arbitrary per-trial equalizer absorb every physiological mismatch. Uncalibrated digital amplitude is not absolute pressure or airflow.

First parameter budget and path to full geometry

Begin with a proposed 6-10 global anatomical degrees of freedom and a small set of pose/source variables per sustained segment. Select the exact subset after examining the simulator and testing sensitivity; do not promise a specific parameter exists in an upstream API. Candidate global dimensions include supported oral/pharyngeal lengths, palate dimensions, lip width, and a nasal dimension only if independently represented. All unestimated properties remain explicit prior assumptions, not recovered anatomy.

The first model is an actual anatomical hypothesis even if low dimensional. Every view and result names the estimated subset. Increase model richness to a deformable 3D anatomical mesh, branched airway acoustics, and coupled mechanics as recovery tests justify doing so. More mesh vertices are not inherently more information. A tube area function alone does not determine a unique 3D cross-section; if the renderer uses circular sections or a template, record that assumption.

For mechanics, start by testing which parameter combinations are distinguishable. For example, oscillation frequency can constrain a stiffness-to-mass relationship without uniquely determining both quantities. If multiple mass/stiffness/pressure combinations explain all available observations, retain them. More recordings cannot resolve a symmetry that every experiment preserves.

Nonlinear source-filter interaction is relevant to phonation, so the long-term system must test a coupled physical model rather than equating filtered audio with complete physiology. High singing pitches also sample resonances sparsely; uncertain spectral peaks should not become exact formant targets.[3][4]

3. Titze. Nonlinear source-filter coupling in phonation, 2008.
4. Bresch and Narayanan. Real-time MRI study of resonance tuning in soprano singing, 2010.

4. Forward simulator is the critical dependency

Use VocalTractLab as the first integration target. It provides an articulatory forward model and native backend. Pin a tested release or commit and speaker asset hashes. The development repository explicitly warns that it includes experimental changes. Build its tests, render a known example, and inspect parameter capabilities before writing the inverse solver.[5]

The documented C interface supports speaker initialization, parameter metadata, tract-to-tube conversion, contour export, and synthesis. Use these to build a narrow adapter. Global morphology changes may require speaker configuration or deeper native integration; they are not interchangeable with frame-level tract controls. Run isolated simulator workers so mutable synthesis state is not shared between candidates.[6]

Create a machine-readable capability manifest: parameter name, unit, anatomical versus pose versus source class, valid bounds, update method, reset requirements, and whether the parameter changes audio, visible geometry, or both. Test each supported control with perturbations. A successful API call is not proof that the control has the intended physical effect.

Nostril occlusion is a special integration gate. Changing velum aperture is not equivalent to blocking the nostril outlet. Inspect and test whether the chosen solver can represent the nasal branch and its outlet impedance. If not, implement and validate that physical boundary condition before using occlusion data in inference. Exclude unsupported trials from the model likelihood; mark them unmodeled rather than silently substituting another control.

A smaller independently implemented branched tube model is a contingency if VTL integration fails. It must conserve the intended flow/pressure relations, pass simple closed/open-boundary and resonance tests, and expose its restricted assumptions. It still attempts inverse physics; replacing it with a cue lookup table is not an acceptable fallback. Record engine substitution prominently because it changes the scientific scope.

Do not assume the native simulator is differentiable. Start with bounded sampling and derivative-free optimization. Later train a differentiable surrogate on simulator outputs for acceleration, always checking selected solutions against the original solver. A surrogate's confident prediction outside its training region is not physical evidence.

5. Inverse fitting and uncertainty

The ideal target is a joint posterior over anatomy, mechanics, articulation and nuisance variables conditioned on all calibration observations and intervention records. The hackathon implementation approximates it with diverse bounded candidates and explicit fit diagnostics. Optimization restarts alone do not constitute calibrated Bayesian uncertainty.

```
For each proposed anatomy theta and mechanics setting phi:
  fit bounded articulation / source / capture variables to calibration data
  simulate the same recordings with shared theta and phi
  score acoustic fit, visible fit, motion constraints and prior plausibility
  keep diverse good explanations, including correlated parameter alternatives

Choose a new experiment using the surviving explanations
Commit predicted outcomes before acquiring its evaluated recording
Score outcome; update the model only after prospective scoring
```

Use joint acoustic and visual residuals with quality-dependent weights. Acoustic features include harmonic magnitudes, robust spectral summaries, F0 and task timing; use reliable resonance estimates where available. Compare rendered audio at multiple time/frequency resolutions rather than relying on one formant extractor. Scale losses by estimated noise and feature correlation; thousands of correlated frames do not imply thousands of independent anatomical measurements.

5. TU Dresden. VocalTractLab development backend and GPL-3.0 LICENSE,
6. TU Dresden. VocalTractLab C API header. Parameter metadata, tract conversion and synthesis interfaces;

For image evidence, fit camera pose and scale explicitly. Without a known reference or independently calibrated geometry, metric dimensions may remain ambiguous. MediaPipe landmarks are useful relative observations, not a direct metric scan. Reprojection must use an actual mapping between model surface points and observed landmarks; missing hidden correspondences contribute no fabricated observations.[7]

Initialize with bounded space-filling samples, then refine several distinct candidates. A first compute target is 256 global samples and 8-16 refinement starts, subject to measured runtime; record the actual budget and convergence. Fit a few stable segments first, add trajectories later. Cache immutable simulations by parameter/configuration hash. Do not silently discard candidates just because a job timed out.

Maintain separate outputs for acoustic residual, visual residual, prior penalty and constraint violation. A good total score can conceal a poor acoustic fit if the prior dominates. Repeat fitting under wider plausible priors, different starts, and altered noise assumptions. Report candidate spread as an approximation until synthetic coverage tests justify probability intervals.

Test local parameter sensitivity using finite differences and inspect nearly dependent response directions. Use profile fits to search for materially different anatomy with almost the same score. Local sensitivity is not a proof of global identifiability; explicit alternative solutions and held-out interventions are essential.

Include a model-mismatch state. If every candidate fails, broaden the physical family or revise the measurement model. Do not force a best candidate to masquerade as a successful reconstruction. A flexible learned residual may improve audio reproduction while destroying anatomical interpretability; compare fits with and without it, bound its capacity, and keep it out of the initial claim.

6. Active physiological experiments

Astra's job is to identify competing physical explanations and propose the next safe, feasible experiment. Numerical simulation supplies predicted observables and disagreement. The planner chooses from validated actions rather than inventing numerical anatomy or treating verbal instructions as perfectly executed motor commands.

The principled objective is expected information gained about shared anatomy/mechanics, marginalized over uncertain articulation and capture, minus participant effort and execution cost. In the prototype, use noise-normalized predictive disagreement among candidate models as a heuristic and label it accordingly. Large disagreement caused solely by uncertain task execution is not useful anatomical information.

Candidate experiment	Intended constraint	Main ambiguity / control
Matched vowels at comfortable pitches	Different shapes must share one underlying anatomy.	Tongue and source may both change; use visible motion and repeats.
Visible change in lip opening or rounding	Constrain the downstream opening and test predicted spectral changes.	Jaw and tongue can compensate; record actual visible movement.
A gentle pitch change on the same vowel	Sample the response at different harmonics.	Tract configuration may change with pitch; do not assume it is constant.
Teacher-reviewed nasal contrast or brief occlusion task	Probe coupling or a modeled outlet boundary.	Outlet occlusion is not velum closure; sound and behavior may both change.
Normal-view front/profile capture	Constrain visible surface morphology and movement.	Camera calibration and scale are uncertain; hidden geometry remains inferred.

7. Google. MediaPipe Face Landmarker web guide, Apache-2.0 source and bundle model cards,

This table defines research candidates, not a self-administered exercise prescription. Use comfortable ordinary phonation and a reviewed protocol with a consenting adult. An occlusion task enters the study only after teacher/voice-specialist review and verification of the solver boundary condition. Do not ask participants to insert devices into their mouths, force high notes, or continue through pain or worsening voice. Stop works locally.[8]

The nasal branch is scientifically motivated: singing studies have combined airflow, imaging and acoustic modeling, and recent speech inversion research uses nasalance information. Neither establishes that a single phone recording measures the connection's exact shape. We will test whether repeated interventions reduce the relevant ambiguity.[9][10]

The intervention execution problem

Define two different prediction tasks. A **prospective task prediction** integrates over how the person may execute a cue, using only information available before recording. A **conditional acoustic prediction** may use measured post-intervention visible movement but must freeze anatomy and exclude that trial's acoustic features from fitting. Report them separately. Fitting the held-out sound's articulators to that same sound is reconstruction, not prospective prediction.

Every action record includes an intended manipulation, permissible variation, expected visible compliance, task/pitch/level context, repetitions, rest/stop rules and a predicted observable distribution. Use matched repeat trials and counterbalanced order where feasible. Manual protocol notes can report deviations. Acquire independent baseline repeats to estimate noise before declaring a small spectral change informative.

Embodied cues and dynamic mapping

The teacher must translate physical hypotheses into familiar actions that a singer can discover, feel, recall and transfer into singing. Examples include a reviewed duck-like sound for a twang-like contrast, a gentle plaintive imitation, noticing comfortable side-rib/back movement, and ordinary vowel/yawn-like/tongue gestures during mapping. These are candidate ways to elicit behavior, not commands that guarantee a specific hidden movement. Yawn-sigh has a clinical facilitating role; neither that precedent nor acoustic twang findings validates every mnemonic or a one-to-one anatomical interpretation.[11][12]

The complete [embodied learning and motion addendum](embodied-learning-and-motion-plan.md), also included in the PDF, defines cue review, learner sensation reports, dynamic capture, motor-control inference and transfer testing. Coughing is retained as a specialist-review candidate rather than an automated exercise; comfortable attempts replace forced maximal inhalation or tongue extension. Self-touch does not verify individual muscle activation, and a cry-like sound does not prove measured laryngeal tilt.[8]

Extend the physical hierarchy with a separate cue-conditioned control profile: observed comfortable envelope, repeatability, timing, context and voluntary recall. Stable anatomy, current articulation, learned control and subjective sensations remain distinct. Never infer an anatomical limit from one failed cue. PRED-01 integrates uncertainty about cue execution; actual measured movement belongs only in separately labeled conditional predictions after capture.

During mapping, record neutral-to-gesture-to-return trajectories with synchronized RGB/depth/audio, head-pose registration and per-frame visibility. Visible soft-palate motion can be estimated only when consistently observed and calibrated; missing depth is not a zero excursion. Report observed movement rather than claiming full physiological range. Multiple moving poses must not be fused as a rigid static cavity. Hidden movement remains an inference target with explicit uncertainty.

Add CUE-01 (B: reviewed cue library and embodied teaching), MOT-01 (A: motor-control model), MOT-02 (B capture/A fusion: dynamic mapping) and LEARN-01 (B: independent recall/transfer evaluation). These extend the prototype now. Later coaching-efficacy studies remain separate from demonstrating a working teaching loop. Acceptance adds G6 repeated dynamic mapping and G7 a measured goal -> cue -> sensation -> recall -> phrase-transfer cycle; synthetic runs cannot establish human learning.

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8. National Institute on Deafness and Other Communication Disorders. "Taking Care of Your Voice." Updated June 11, 2025.
9. Gramming, Nord, Sundberg and Elliot. Does the nose resonate during singing? KTH STL-QPSR 34(4), 1993.
10. Tabatabaee et al. Enhancing Acoustic-to-Articulatory Speech Inversion by Incorporating Nasality. Interspeech 2025.
11. American Speech-Language-Hearing Association. Voice Disorders; yawn-sigh and facilitating approaches.
12. Sundberg and Thalén. What is Twang? Journal of Voice, 2010. Single-performer acoustic/source study.

7. Learning a population prior from video and other data

The proposed population model learns relationships among physical parameters, visible articulation and sound, then initializes individual inference. Train it explicitly; merely placing examples in Astra's context does not update a persistent model's weights. Astra can help specify features, critique failure cases and propose experiments, while dedicated audio/vision models and the physics solver provide observations and numerical inference.

Synthetic physics data first. Sample multiple plausible anatomical templates, articulation trajectories, source settings and capture conditions. Store exact generating parameters, solver version and parameter coverage. This provides labeled inverse problems immediately. It also imports the simulator's biases; success on the same generator only verifies that model family.

Instrumented audiovisual data next. Seek paired audio and modality-specific MRI, ultrasound, EMA or other measured articulation with compatible reuse permissions. Use each measurement only for the variables it can validate. Ordinary speech differs from singing, and data acquired in a scanner may differ from natural performance. Do not assert that a convenient dataset contains singing, 3D anatomy, tissue mechanics and aligned face video unless verified.

Permitted singer videos as weak supervision. Prioritize synchronized, unprocessed solo demonstrations, especially repeated changes within one singer. Extract lip aperture/rounding, head pose, visible jaw motion, timing and acoustic features. Audiovisual inversion research supports the usefulness of facial movement; this does not prove external nose dimensions identify internal nasal geometry.[13]

Static facial proportions are candidate features, not assumed anatomical truth. Test their incremental benefit beyond dynamic landmarks and audio. Hold out singers, channels and sessions; identity, language, style, microphone and processing can otherwise masquerade as anatomical relationships. Audio/video mismatch, dubbing, accompaniment and enhancement require quality flags. Source separation artifacts are additional uncertainty, not clean physiological evidence.

Internet video alone supplies no ground-truth hidden anatomy. A network trained only for audiovisual reconstruction can learn useful associations without discovering correct internal structure. Anchor it with physical constraints and independent measurements, and compare to audio-only and shuffled/static-feature controls. Never treat Astra-generated anatomical labels as independent ground truth.

For the first implementation, skip large-scale internet collection and training. Use a generic physics prior, generate a bounded synthetic bank, and run live individual fitting. This keeps personal anatomy inference central. Population training is the next acceleration/generalization phase, with a manifest documenting permission, model/data terms and permitted redistribution for every asset.

8. Application architecture and Astra tools

13. Engwall. Audiovisual-to-articulatory inversion. Speech Communication 51, 2009, pages 195-209.

```
Native iPhone RGB / depth / audio capture + web display
          |
          v
Timestamped evidence bundle + consent + task execution record
          |
          v
Python analysis -> inverse solver -> isolated VTL native workers
                         |                    |
                         v                    v
                  candidate model store <- simulation cache
                         |
                         v
               numerical experiment evaluator
                         |
                         v
               Astra hypothesis / action planner
                         |
                 validated experiment proposal
                         |
           committed prediction -> next capture -> evaluation
```

Use AudioWorklet for bounded capture and a Worker for analysis; server fitting is asynchronous. Preserve original capture settings and timestamps, disable enhancement where supported, detect clipping/noise/dropouts, and recalibrate after device changes. Local pitch/quality feedback remains useful instrumentation, not the primary result. A few stable vowel segments are enough to exercise the inference pipeline; a full lesson UI is unnecessary.[14][15]

Expose typed tools: `inspect_evidence`, `fit_person_model`, `compare_candidate_models`, `simulate_experiment`, `rank_experiments`, `commit_prediction`, and `explain_result`. Each response includes evidence IDs, model version, numerical diagnostics, units and invalidity reasons. The host validates proposals and runs the actual software. Astra cannot overwrite measurements or promote uncertainty labels by narration.

The currently documented Astra API accepts text and images rather than native audio/video. Give it structured measurements, selected frames and plots, while specialized tools analyze audio and sequences. This is a real integration constraint, not a change in the moonshot. Benchmark the full system against fixed and simpler planners to test whether Astra improves experiment efficiency; exclusivity is not assumed.[16]

Prototype stack: native Swift iPhone capture companion plus React/TypeScript/Vite display client; Python analysis and inverse service; native simulator adapter; SQLite plus local artifact files for a supervised single-user experiment. Production adds authenticated Postgres, explicit media retention, isolated job queues and object storage. No API key in the client. Log solver and model versions, seeds, action budgets and failures so another team can reproduce the inference.

Proposed performance budgets: local capture feedback under 150 ms p95 on the chosen device; first inverse update within 30-120 seconds for short segments on measured hardware; next-trial planning after numerical results. These are targets to benchmark, not guarantees. Cache and background refinement can improve interaction. Do not put a full anatomical optimization inside the audio callback or promise real-time tissue reconstruction.

Integrated iPhone capture and depth workstream

Acquisition owner: Person B. Scientific fusion owner: Person A. Repository integration owner: Person B. B owns microphone, RGB and depth together, including timestamps, calibration, quality, native deployment and export. A owns the joint observation model, depth projection, visibility masking and how these measurements constrain anatomy. Both review the interface and acceptance results. This is part of the main build, not an unassigned future extension.

Available devices are an iPhone 13 Pro Max and iPhone 15 Pro; both provide front TrueDepth and rear LiDAR. Start by testing the 15 Pro front TrueDepth and microphone together, then compare the other device and rear mode on the

14. Mozilla Developer Network. "AudioWorklet." Official browser API reference,
15. Mozilla Developer Network. "MediaDevices: getUserMedia() method" and capture constraints.
16. OpenAI. "GPT-6 Astra Model." Modalities, tools and pricing,
17. Apple. iPhone 13 Pro Max and iPhone 15 Pro technical specifications.

same targets. Capture format availability is checked at runtime. Use a Mac for compute/display. Do not assume front and rear depth can run concurrently.[17]

B builds a native Swift capture companion. Use supported AVFoundation video/depth/audio outputs and time-matched delivery, preserve source timestamps and verify alignment. If an ARKit capture mode is used, explicitly align its timestamps to audio. A web camera feed is not a substitute for the depth and calibration payload. The first integration uses a short recorded bundle; network streaming follows after correctness.[18]

Define a versioned ObservationBundle together before independent implementation: trial and prediction IDs, audio samples/timestamps, RGB frames/timestamps, depth samples with their own timestamps, depth/disparity representation and units, intrinsics and alignment transforms, capture settings, confidence/validity, dropped frames, filtering flags and artifact hashes. Files may be separate, but share a manifest and timebase. Network arrival time is not capture time. A consumes all modalities jointly, not unrelated summaries.

Depth directly constrains only visible surfaces. Benchmark known-size and cavity-like bench objects before close-range human capture. Measure bias, repeatability, missing samples and alignment error. Test visible face/mouth surfaces without assuming that a rendered face mesh measures the oral interior. Hidden nasal/throat structures remain physiological inference targets. Do not fill depth holes and call them measured anatomy; preserve less-filtered data where supported.[19]

Depth milestones and acceptance ownership

Ticket	Owner	Acceptance evidence
CAP-01	B; A reviews schema	App deployed to a phone; actual available formats logged; short RGB/depth/audio capture or precise blocker recorded.
CAP-02	B	Exported bundle replays with timestamps, calibration, missing-data flags and measured synchronization error; bench-target quality report.
FUSE-01	A; B supplies bundle and test	One real bundle enters the inverse pipeline; audio, RGB and valid depth influence the fit; missing depth is handled explicitly.
DEPTH-01	A numerical forecasts; B execution/evaluation	Prediction frozen before acquisition, followed by scoring and then update; repeat capture uses the same contract.
DEPTH-02	B runs evaluator; A reviews fits	Audio-only, audio+RGB and audio+RGB+depth compared under equal budgets; effect on prediction/recovery reported, including null results.

At each capture and integration gate, both people run the shared artifact through their side of the interface. Exchange exact commit, command, contract version, tests and blockers. If native deployment fails, B assigns a bounded diagnosis task and maintains synchronized browser RGB/audio as the working capture path. Depth stays explicitly unfinished; it is not silently replaced with landmark depth. A continues synthetic inverse-physics work while B resolves capture. Reduce visual polish before dropping these integration checkpoints.

Person B's additional native work is offset by keeping the interface minimal. Person A takes numerical calibration checks and depth residual implementation. Simultaneous two-phone capture is an extension after one-phone

17. Apple. iPhone 13 Pro Max and iPhone 15 Pro technical specifications.
18. Apple. AVCaptureDataOutputSynchronizer and front TrueDepth capture timestamps.
19. Apple. Discover advancements in iOS camera capture: Depth, focus, and multitasking. WWDC22; depth filtering guidance.

synchronization works, because it adds clock drift and spatial registration. Static scans and dynamic singing remain distinct observations, not one rigid reconstruction.

9. Data contracts, visualizations and reliability

Record	Required content
EvidenceBundle	Participant/session IDs; consent scope; device/settings; timestamps; raw-asset hashes; features and units; quality/missingness; camera calibration; task execution; synchronized depth and its validity/calibration metadata.
ParameterManifest	Physical meaning; units/bounds; global versus dynamic role; simulator mapping; measured/inferred/fixed status; supported manipulation.
PersonModelVersion	Parent version; all fit evidence IDs; global candidate parameters; dynamic states; nuisance fit; residuals; priors; solver/config hashes; uncertainty method and mismatch flag.
ExperimentSpec	Intended intervention; feasibility/review status; target context; allowed variation; expected compliance observations; stop rules; selection rationale.
PredictionCommit	Immutable model/evidence IDs; action; predicted observables and uncertainty; scoring rule; time; evaluation mode; hash committed before capture.
EvaluationRecord	Actual execution; prediction ID; eligible observations; per-metric errors; exclusions; baseline comparison; no post-outcome edits to prediction.

Implement a state machine: calibration -> fit -> proposed -> committed -> capture -> score -> update. Recording before commitment is unpredicted data and cannot enter prospective scoring. Missing or poorly executed trials remain counted with reasons; they are not silently dropped from the denominator. Reject stale results after stop, consent changes or model revision. Cancel or quarantine pending jobs; retries must be idempotent.

Display an actual parameterized anatomical hypothesis, rendered from the same configuration used for acoustic simulation. Offer a sagittal view, 3D geometry where available, predicted versus observed spectra, and a comparison of competing configurations. Color distinguish visible measurements, inferred geometry and fixed prior assumptions. If two internal shapes explain the data, show both instead of smoothing them into one unsupported scan.

Let the user select a parameter and hear/see a simulated perturbation while keeping its hypothesis status clear. If the renderer only uses an area-function mesh, say how cross-sectional shape was assumed. Never produce an independently generated anatomical illustration and imply it is the fitted solver geometry.

Raw media stays local by default in the hackathon. Obtain separate consent for server transfer, retention and research reuse when needed. Treat inferred physiology as sensitive personal data. Revoke access and delete raw and derived artifacts together when requested; account for queued jobs and backups in production. Images, lyrics and retrieved materials are untrusted data, not instructions to the planner.

10. Open-source components and licensing

This project will be open source. Select components primarily for scientific suitability, accuracy and integration effort; GPL and AGPL components are eligible. Recommend AGPL-3.0 for original networked application code if hosted modifications should remain available to users, or GPL-3.0 if redistribution reciprocity is sufficient. Finalize the exact license identifier and notices at implementation kickoff. Model weights, datasets, inherited source and visual/audio assets need separate provenance records.[20]

GPLv3 and AGPLv3 explicitly permit combinations under their Section 13 provisions while preserving the licenses of the respective parts; the AGPL network-source requirements apply to the combination. Apache-2.0 is compatible with GPLv3. This does not permit relabeling all third-party code or removing separate data/model restrictions. Open-source licenses allow commercial use; a noncommercial restriction is a different category.[20]

Component	Role	License / recommendation
Pitchy	Browser F0 and clarity; clean monophonic baseline/fallback.	Actual LICENSE is 0BSD. Start here; benchmark on intended singing and devices.[21]
librosa	Python phrase DSP, spectral features, alignment and evaluation.	ISC. Use outside the real-time render callback.[22]
SwiftF0	Compact ONNX pitch estimator; browser/WASM candidate.	MIT repository and bundled model. Promising author benchmarks; verify receptive field and streaming behavior.[23]
torchcrepe / CREPE	Independent pitch benchmark and optional phrase corroboration.	MIT code; track converted checkpoint provenance. Silence and sequence-decoding behavior need care.[24]
MediaPipe Face Landmarker	Optional visible lip/jaw/head observations.	Apache-2.0 code and reviewed bundle cards. Relative face depth and blendshapes are not hidden anatomy or calibrated joint angles.[7]
Pink Trombone / modular fork	Small forward-simulation reference or UI comparison; not a substitute for the fitted physical model.	MIT. The modular fork uses AudioWorklet; verify Safari and preserve inherited notices.[25]
Three.js	Renderer for fitted geometry and alternative anatomical hypotheses.	MIT; imported anatomical assets require their own rights. Use simulator-derived geometry; 2D contours can precede a 3D export.[26]
WORLD / pyworld	Offline acoustic analysis/resynthesis research.	BSD-style core / MIT wrapper. Spectral envelope does not uniquely identify anatomy.[27]
Basic Pitch	Later reference transcription / note events.	Apache-2.0. Not the continuous low-latency cents/vibrato signal.[28]
VocalTractLab	Primary hackathon forward engine for physiology inference.	GPL-3.0 is eligible for this open-source project. Pin engine/assets and validate the parameter adapter; integration is now on the initial critical path.[5]

5. TU Dresden. VocalTractLab development backend and GPL-3.0 LICENSE,
7. Google. MediaPipe Face Landmarker web guide, Apache-2.0 source and...
20. Free Software Foundation. GNU license guidance and GPLv3/AGPLv3...
21. Ian Prime. Pitchy repository and actual BSD-0-Clause LICENSE,
22. librosa project. Repository and ISC LICENSE,
23. SwiftF0 project. Repository, MIT LICENSE, bundled ONNX model and 2...
24. Morrison et al. torchcrepe and CREPE repositories, MIT licenses and ch...
25. Thapen et al. Pink Trombone source mirror and modular AudioWorklet fo...
26. Three.js project. MIT LICENSE,
27. WORLD / pyworld projects. Core BSD-style license and MIT wrapper,
28. Spotify. Basic Pitch repository and Apache-2.0 LICENSE,

Component	Role	License / recommendation
Praat / Parselmouth	Reference phonetic analysis and potentially production phrase analysis.	GPL variants are eligible. Select according to singing-metric validity, runtime and reproducibility rather than a proprietary-license concern.[29]
VocalTrax	Promising JAX inverse-acoustics prototype.	MIT top level, but inherited upstream code has unresolved licensing. Do not adopt into product until provenance is resolved.[30]

SwiftF0 reports a 16 ms output hop and a compact model. Those are not demonstrated microphone-to-feedback latency. Conduct a bake-off against Pitchy and torchcrepe, select one production estimator and one fallback, and avoid running all models continuously on every phone.[23]

VocalTrax explicitly credits substantial inherited code from vocal-tract-grad, whose repository lacked a license grant in the inspected root and relevant headers. That is a concrete unresolved provenance issue, not a finding of infringement. Obtain clarification or independently implement published equations using licensed components.[30]

Essentia's AGPL code is an eligible option for the open-source project; add it if its broader DSP functionality beats the smaller initial stack. Its pretrained models have separate noncommercial/no-derivatives terms. openSMILE's research license restricts commercial/product use and is not made unrestricted by open-sourcing this app. Obtain applicable permissions or avoid restricted assets. GPL and AGPL permit commercial use subject to their obligations.[31]

VocalSet's publisher metadata specifies CC BY 4.0 and provides singing-technique material suitable for acoustic testing. It contains no measured individual anatomy or coaching outcomes. GTSinger offers richer technique/multilingual material but carries CC BY-NC-SA 4.0 and additional agreement text; do not assume permission for commercial use.[32]

The distinctive training/evaluation corpus must be collected with consent: calibrated observations, assigned intervention, actual execution, prior model and committed prediction, plus independent physiology where available. Split by singer/session/song as appropriate; adjacent frames or clips from the same recording must not leak across evaluation boundaries. Dataset reuse permission does not remove all performer-consent or downstream-purpose questions.

11. Parallel-agent implementation responsibilities

Two human leads coordinate bounded Astra agents through dependency gates, without fixed hackathon timing. A owns scientific interpretation and numerical acceptance; B owns the experimental system, independent evaluation and repository integration queue. The companion [parallel-agent execution plan](two-person-execution-plan.md) defines exclusive module ownership, dispatch rules, acceptance experiments and human checklists. Its complete text is also included as the execution appendix in the PDF.

Start with KIT-01: a runnable contract kit defining observations, model candidates, forecasts, immutable prediction commits, evaluation records, units, coordinate frames and evidence provenance. Publish schemas and validation commands first; add a genuine synthesized audio/geometry pair and actual synchronized phone capture as their producers become available. The engine capability manifest evolves from verified support. No production consumer uses fabricated fitted anatomy.

23. SwiftF0 project. Repository, MIT LICENSE, bundled ONNX model and 2025 paper,
29. Praat / Parselmouth projects. GPL license evidence,
30. PapayaResearch. VocalTrax README / MIT LICENSE and credited vocal-tract-grad upstream,
31. audEERING openSMILE Research License; Essentia licensing information,
32. Wilkins et al. VocalSet 1.2, publisher record and CC BY 4.0 metadata; GTSinger dataset license,

Agent lane	Human lead	Deliverable / dependency
KIT-01 / INT-01 contracts and integration	B	Versioned artifacts, validation, replay and cross-service checks; minimum shared decisions first.
PHY-01 forward physics	A	Pinned engine, parameter capabilities and real audio/geometry export.
CAP-01/02 native acquisition	B	Synchronized phone audio/RGB/depth, calibration and measured quality; bundle schema first.
AUD-01 acoustic measurements	B; A reviews	Canonical extractor shared by simulated and observed signals; feature definitions first.
GEO-01 visual/depth measurements	A	Calibrated visible-surface constraints, uncertainty and missing data; bundle coordinates first.
INV-01 / FUSE-01 inverse inference	A	Shared anatomy, dynamic articulation, diverse candidates and multimodal fit; engine and measurements first.
SVC-01 scientific jobs	A	Artifact/job lifecycle, cancellation, retries and model version checks; contracts first.
EVAL-01 / DEPTH-02 / REP-01 evaluation	B	Hidden scoring truth, baselines, modality comparisons and reproducibility; frozen scoring protocol first.
PRED-01 numerical forecasts	A	Supported interventions and numerical disagreement; engine capabilities and candidate models first.
ACT-01 / DEPTH-01 orchestration	B	Validated action, frozen prediction, capture, score, then update; forecast/capture/evaluation contracts first.
VIS-01 model visualization	B	Actual engine geometry and alternatives; genuine geometry export first.

Activate roughly six foundation tasks initially: contracts, engine, native capture, acoustic measurement definition, evaluation design and visual/depth feasibility. Start each independent portion once its inputs exist. After genuine artifacts arrive, activate inverse fitting, job service, visualization and executable benchmarks. Then connect numerical forecasts and orchestration into the prospective loop and run modality comparisons. These are dependency waves, not group-wide barriers: synthetic recovery continues while phone capture is unresolved, and capture continues while inference is incomplete.

B integrates reviewed commits in dependency order and maintains a passing cross-service path. Each implementation agent has an isolated worktree, bounded task, exclusive owned files, starting commit, input contract and acceptance experiment. Assign one writer per shared schema/dependency file. A signs off scientific changes; a separate reviewer checks evidence. Every handoff includes actual test results, exact replay commands, provenance and blockers. Do not start new tasks merely to occupy agents or let agents expand their own scope.

Integration evidence gates are G0 contract skeleton; G1 genuine artifact exchange; G2 hidden synthetic inference and actual geometry rendering; G3 real multimodal fitting; G4 prediction committed before capture and scored before update; G5 reproduced benchmarks and modality results. G2 can precede phone capture; synthetic G4 does not satisfy a human-data gate. Task status is proposed, ready, running, review, integrated or blocked, with an explicit missing dependency and owner.

Use additional agents for bounded identifiability, anatomy/articulation separation, microphone compensation, model-mismatch, depth-value and leakage investigations. Algorithm alternatives share an interface and predeclared development benchmark/compute budget; select on development cases and evaluate the selected method on fresh held-out cases. Keep scoring truth outside inference and planning inputs. Agent agreement is not empirical evidence. Schedule simulator/device resources and isolate stateful engine workers when thread safety is unproven.

The demo should show: initial candidate anatomies, why a specific new experiment distinguishes them, the sound/spectrum predicted before recording, the observed result, and how the candidates change. Follow with recovery plots on hidden synthetic subjects. If a compatible independently measured case exists, present that as a separate anatomy validation panel.

Fallbacks preserve the research question. A replayed prerecorded run is labeled as replay. A synthetic-only run is labeled as synthetic. A low-dimensional solver states its variables. No ground-truth synthetic anatomy enters the inverse algorithm, Astra context, or selection logic. No prerecorded success is presented as a live reconstruction.

Implementation tickets and acceptance criteria

1. **PHY-01 forward engine:** reproducible build; unit-aware adapter; reset tests; supported boundary conditions; known configuration yields deterministic audio/geometry.
2. **INV-01 shared anatomy:** fit several tasks jointly; parameter domains enforced; multiple starts; synthetic truth withheld; fixed-anatomy baseline using equal dynamic flexibility.
3. **OBS-01 capture (B):** includes CAP-01 and CAP-02 native RGB/depth/audio capture; synchronized timestamps, calibration/quality, permission loss and local stop. **FUSE-01 (A)** consumes the bundle; **DEPTH-01/02 (joint)** test prospective use and incremental value, as specified in section 8.
4. **ACT-01 experiment planning:** numerical forecasts for available actions; compliance uncertainty; fixed-protocol baseline; Astra rationale references actual candidate evidence.
5. **EVAL-01 prospective record:** immutable prediction precedes capture; scoring precedes update; no target leakage; all failures/attempts retained in summary.
6. **VIS-01 anatomy display:** geometry derived from fitted engine; alternative candidates visible; fixed versus estimated parameters clear; predicted/observed differences readable.
7. **REP-01 reproducibility:** one command reproduces synthetic benchmark and recorded fit from a permitted manifest; versions, seeds and budgets included.

12. Validation that can disprove our model

For simulation, hold out entire anatomical instances and intervention sequences, not adjacent frames. Measure parameter error normalized to each physical range, recovery of identifiable parameter combinations, predictive error, and interval coverage where intervals are claimed. Include noisy capture and mismatched models: altered wall losses, room response, or a richer generating model than the inverse solver. Same-simulator recovery is vulnerable to an inverse crime and cannot establish real anatomy.

For live experiments, lock feature/scoring definitions before outcomes. Evaluate task predictions without refitting on the held-out sound. Report recording quality, attempt count, exclusions, runtime and comparison budgets. Compare intervention effects relative to repeated-baseline variability, not just whether a generated voice sounds similar. Very small hackathon samples are case studies, not population efficacy evidence.

For anatomical validation, use withheld modality-specific measurements, registered into a declared coordinate system. Static MRI may constrain geometry in the imaged pose; dynamic imaging constrains observed trajectories. EMA covers tracked points, not complete volume. EGG provides contact-related information, not a direct stiffness or mass measurement. Instrumental assessment must be conducted through appropriate research/clinical partners.[33][34]

Separate training data, subject calibration data and scoring-only measurements. If an image of the target's internal tract initializes their reconstruction, label that MRI-assisted personalization; it does not validate phone-only inference. For mechanics, first establish recoverability in controlled physical or synthetic systems with known material parameters, then seek suitable independent human evidence. Geometry agreement alone does not prove tissue mechanics.

Predeclare parameter-specific tolerances according to measurement resolution and intended use once a reference dataset is selected; do not invent a universal millimeter threshold before knowing the reference. Compare anatomical error against a generic template and plausible-prior baseline, including uncertainty. A posterior that confidently excludes the reference is worse than an appropriately broad one.

Failure pattern	Interpretation / next experiment
Great audio fit, wrong reference geometry	Acoustic fitting succeeded; physiological reconstruction failed. Test alternatives and stronger measurements.
Anatomy changes across vowels or microphones	Global/dynamic/capture separation failed; audit model freedom and calibration.
Active actions add no information	Selection may target noise or redundant observations; compare repeat noise and parameter sensitivity.
Predictions fail under new interventions	Overfitting or missing physics; inspect execution uncertainty and solver boundaries.
Face features help only on familiar singers	Possible identity/capture confounding; enforce singer/source holdouts.
Candidate spread collapses without accuracy	False confidence from likelihood/prior/model mismatch; test coverage and alternate model families.

13. Research-to-production roadmap

Reproducible inverse-physics baseline. Stabilize the supported anatomical subset, synthetic benchmarks, inference diagnostics and experiment state machine. Benchmark engine runtime and test several consenting cases under a reviewed protocol. Gate: repeatable fitting and honest mismatch detection, not a predetermined positive result.

Active multimodal inference. Add a validated nasal boundary model if missing, more visible constraints, richer global morphology and a simulation surrogate where useful. Compare active, fixed, audio-only and audiovisual conditions at equal budgets. Gate: reproducible predictive improvement and reduced error/ambiguity in known-parameter settings.

33. American Speech-Language-Hearing Association. "Vocal Tract Visualization and Imaging."
34. Herbst. "Electroglottography - An Update." Published online March 11, 2019; journal issue July 2020.

Independent anatomy study. Secure partners and modality-compatible data, define anatomical targets and reference registration, and evaluate unseen people. Instrument access, permissions and data quality are prerequisites. Gate: parameter-specific reference agreement beyond generic templates; no promise this will succeed.

Following phase, mechanics and fuller 3D inference. Add self-oscillating source/tissue models, coupled dynamics and more expressive geometry. Run identifiability and mismatch tests before attributing fitted constants to physical tissues. Gate: independent evidence for each new physical interpretation, not just lower acoustic error.

Broader coaching validation after the prototype embodied loop. Use simulated interventions to guide comfortable practice, then evaluate learning, retention and usability. The production roadmap adds identity/access controls, deletion, device reliability, specialist-reviewed exercises, monitoring and model drift handling. A scientifically inconclusive model may remain a valuable open-source research tool without being advertised as an accurate personal physiological coach.

14. Decisions, remaining unknowns and deliverables

The first dispatch establishes KIT-01 and launches the independent foundation tasks in section 11, including cue-library/protocol design and synthetic motor-control design from the embodied addendum. PHY-01 and the synthetic recovery harness form the scientific critical path while native capture, measurement validation and evaluation design progress alongside them. Pin the engine and supported parameter subset, generate hidden test anatomies, and attempt reconstruction from simulated audiovisual tasks. This tests the coherence of the inverse loop while the experimental instrument is being validated.

Critical unknowns are supported nasal outlet control, global morphology integration, runtime per candidate, correspondence between external and internal geometry, robustness to source/capture compensation, and recoverability of mechanics. Each has an explicit experiment or integration gate above. Lack of certainty is expected; hidden substitution of a simpler product is not.

Deliver an open-source research repository with the forward adapter, inverse solver, experiment planner, capture client, actual model visualization, reproducible benchmarks, provenance manifests and a results report. Publish code and appropriately permitted synthetic/sample assets; real participant data requires explicit release permission. This specification and its PDF are design artifacts only. No recovery accuracy, speed or anatomical validity is claimed as achieved.

The intended breakthrough is **recovering an individual's physical voice-generating system through conversation-guided experiments**. The standard for the hackathon is a genuine attempt that could fail in an informative way. The standard for eventually claiming reconstruction is independent evidence that the inferred physiology corresponds to the person.

Execution appendix: parallel-agent plan

Companion to revision 6 of the physiological acoustic moonshot specification. Two humans remain accountable, each coordinating as many bounded Astra tasks as the dependency graph supports. This is an implementation plan, not evidence that the application or anatomical recovery works. There are no fixed hackathon time estimates.

Optimize for the earliest integrated, independently evaluated physiological inference experiment. Launch an agent only when its inputs, owned files and acceptance experiment are concrete. Additional capacity should remove a demonstrated bottleneck or challenge a scientific assumption. Unlimited agents do not imply unlimited simultaneous edits or benchmark compute.

1. Human accountability and coordination

Responsibility	Accountable lead	Review
Forward physics, parameter meanings, inverse inference, visual/depth likelihood and numerical forecasts	A: scientific model lead	B checks evidence and integration
Native acquisition, canonical acoustic measurements, application orchestration, visualization and independent evaluation	B: experimental system lead	A checks scientific validity
Integration queue, shared contracts, cross-service checks and release readiness	B: integration lead	A signs off on scientific changes
Objective, scoring rules, supported anatomy and scope changes	A and B jointly	Record decision before dependent implementation

Each human keeps one coordinating conversation and delegates bounded work to separate agent conversations. Coordinators manage task dependencies, review evidence and resolve decisions; implementation agents return small tested commits. The Astra runtime selecting vocal experiments inside the product is separate from these development agents.

B owns repository integration; A owns scientific inference integration (FUSE-01). This distinction replaces ambiguous uses of "integration owner". No agent merges its own work solely on its own completion claim. Participant interaction, device access and anatomical interpretation remain human responsibilities.

2. First shared deliverable: runnable integration kit (KIT-01)

Agree a narrow first experiment: several comfortable vowel segments constrain one shared anatomical parameter set, followed by a held-out prediction. Nasal occlusion is added only after the engine boundary condition and execution protocol are demonstrated. This preserves the reconstruction moonshot without inventing supported controls.

- [ ] Version observation, candidate-model, forecast, prediction-commit and evaluation schemas.
- [ ] Define units, coordinate frames, time origins, quality flags, artifact hashes, missing-data reasons, job states and model/evidence IDs.
- [ ] Define forward-engine interfaces and an extensible capability manifest; unsupported parameters stay unsupported until verified.
- [ ] Agree canonical acoustic features, calibration/held-out roles, scoring rules and a fixed-anatomy baseline with equal articulation freedom.
- [ ] Publish validation and replay commands with expected outputs.

- [ ] Add one genuine engine-generated audio/geometry pair with provenance as soon as PHY-01 exports it.
- [ ] Add one actual synchronized phone bundle when CAP-01/02 produces it.

KIT-01 has incremental handoffs: the schema skeleton unblocks independent foundations; genuine artifacts unlock the consumers' integration checks. Do not wait for full anatomy discovery to build capture. Development examples must identify their provenance and cannot be reported as live human reconstruction. No production path returns fabricated fitted anatomy.

Start each task from a recorded integration commit. Shared contracts describe file-backed immutable artifacts first; the job API transports the same records. Large media and simulator states stay outside agent prompts.

3. Agent lanes and exclusive module ownership

Paths below are proposed application paths, not claims about files already implemented. Adapt them to existing work with an explicit ownership map before launch. One active writer owns each file, including shared configuration.

Lane / tickets	Lead	Owned paths	Deliverable and acceptance experiment	Prerequisites
Contracts and integration / KIT-01, INT-01	B	contracts/, tests/integration/, repository CI	Validate genuine producer artifacts; replay one cross-service flow; reject incompatible versions and stale results.	Joint minimum interface decisions
Forward physics / PHY-01	A	science/forward/, tests/science/forward/	Pinned build; parameter units/bounds; repeatable synthesis and matching geometry; reset and unsupported-control tests.	Engine selected
Native acquisition / CAP-01, CAP-02	B	apps/ios/, capture/	Deploy on available iPhone; export/replay audio, RGB and supported depth; measure timing/depth quality and missing samples.	Observation schema; human device access
Acoustic measurements / AUD-01	B, A reviews	observations/audio/	One versioned extractor for observed and synthesized signals; controlled pitch/spectral cases, noise and failure behavior.	Feature definitions; real or synthesized test signals

Lane / tickets	Lead	Owned paths	Deliverable and acceptance experiment	Prerequisites
Visual/depth measurements / GEO-01	A	observations/geometry/, tests/science/geometry/	Calibrated visible-surface measurements and uncertainty; known-target projection, masking and missing-depth tests.	Bundle coordinates; actual depth data for device acceptance
Inverse inference and fusion / INV-01, FUSE-01	A	science/inverse/, tests/science/inverse/	Shared anatomy across tasks; dynamic states; diverse candidates; held-out synthetic recovery; real multimodal fit and missing-modality behavior.	PHY-01, AUD-01; GEO-01 for visual/depth fusion
Scientific job service / SVC-01	A	science/service/, tests/service/	Submit, cancel and replay real jobs; immutable artifacts; idempotent retries; stale-model rejection.	Job/artifact contracts; engine for executable integration
Independent evaluation / EVAL-01, DEPTH-02, REP-01	B	evaluation/, tests/evaluation/	Scoring-only truth; equal-budget baselines; frozen forecasts; modality comparisons; reproducible report including failed attempts.	Contract/scoring agreement; executable inputs as producers land
Numerical experiment forecasts / PRED-01	A	science/prediction/, tests/science/prediction/	Forecast supported interventions from frozen candidates; quantify disagreement and compare with repeat variability.	PHY-01 capability manifest; candidate models from INV-01
Experiment orchestration / ACT-01, DEPTH-01	B	experiment/, tests/experiment/	Validate proposals; commit prediction before capture; score before update; stop/failure/retry behavior.	Forecast contracts; PRED-01, capture and evaluator for full loop

Lane / tickets	Lead	Owned paths	Deliverable and acceptance experiment	Prerequisites
Actual model visualization / VIS-01	B	apps/web/	Render engine geometry and alternatives; distinguish fitted/fixed/unsupported structure; align predicted and observed results.	Genuine geometry export and model contract

A owns scientific dependency files and native-engine build configuration. B owns repository CI and frontend/iOS dependency files. If Python dependencies are shared, A is the sole lock-file writer; B requests additions. Reviewers propose changes without concurrently editing an implementer's files. Task transfers are recorded before the new writer starts.

Preserve synchronized evidence while splitting processing

B's acquisition agent owns microphone, RGB and depth together, preserving a shared observation manifest, source timestamps, calibration and synchronization uncertainty. Audio and geometry agents may process independently, but every output references the same evidence IDs/timebase. A's inverse agent combines their constraints in one physiological fit. Never treat independent network arrival times as synchronization or derived face geometry as measured hidden anatomy.

The available iPhone 13 Pro Max and iPhone 15 Pro remain the hardware targets. Start with one-phone synchronized capture; retain CAP-01/02, FUSE-01 and DEPTH-01/02 from the capture addendum. Sensor limitations remain measured outcomes, not reasons to silently remove depth from the work queue.

Embodied learning extensions (revision 6)

The [embodied learning and motion plan](embodied-learning-and-motion-plan.md) is part of this execution plan. Add two independent agent lanes: B owns CUE-01 in experiment/cues/, apps/web/coach/ and tests/cues/; A owns MOT-01 in science/control/ and tests/science/control/. Narrow the existing experiment orchestration owner to workflow/tool execution outside experiment/cues/, and the visualization owner to web files outside apps/web/coach/. The existing capture, geometry and inverse owners implement MOT-02 within their modules; the evaluator owns LEARN-01 in evaluation/learning/. No shared file has two writers.

CUE-01 library/contract design and MOT-01 synthetic control-model design can run with wave 1. Dynamic capture/fusion requires CAP-02, GEO-01 and INV-01; integrated teaching requires PRED-01 and ACT-01. KIT-01 includes CueDefinition, CueAttempt, SensationReport, MotionObservation, ControlProfile and TransferEvaluation. G6 verifies dynamic mapping; G7 verifies a measured cue-to-recall/phrase-transfer loop. See the addendum for exact acceptance evidence and review status of example cues.

A reviews the separation of anatomy, articulation and learned control; B secures teacher review of cues and evaluates recall/transfer independently. Both keep observed movement, hidden-motion hypotheses and self-reported sensations distinct. These are initial implementation responsibilities, not an unowned production-roadmap extension.

4. Dependency-driven activation

These waves describe prerequisites, not dates or mandatory group-wide barriers. A lane starts its next task as soon as its own inputs pass, without waiting for unrelated lanes.

Wave 1: foundations and uncertainty removal

Start roughly six concrete tasks: KIT-01 contracts; PHY-01 engine; CAP-01 native capture; AUD-01 measurement definition/controlled tests; EVAL-01 scoring and leakage harness design; GEO-01 coordinate/measurement feasibility. Capture starts after the minimal bundle agreement; evaluation can specify scores before the engine is available, but cannot claim executable recovery results yet.

A checks what anatomy and interventions the engine actually exposes. B obtains actual device-format logs and recordings. Both freeze the first calibration/held-out protocol. Return runnable evidence for capability claims instead of broad research summaries.

Wave 2: inference and application integration

Once genuine artifacts exist, activate INV-01, SVC-01 and VIS-01. AUD-01 validates engine outputs and live recordings; GEO-01 validates actual depth and produces constraints for FUSE-01. The evaluator generates fresh hidden scoring cases and runs executable baselines. Native capture proceeds independently of synthetic inverse recovery.

Wave 3: prospective experiments and comparisons

PRED-01 consumes real candidate models. ACT-01 connects forecasts to the learner and immutable ledger. EVAL-01 scores outcomes before updates. Run DEPTH-01 followed by DEPTH-02 modality comparisons with equal budgets. Investigate failures with bounded research tasks rather than broad unplanned rewrites.

Critical dependency chains

- Scientific: PHY-01 + AUD-01 -> INV-01 -> PRED-01 -> prospective loop.
- Measured multimodal: CAP-01 -> CAP-02 + GEO-01 -> FUSE-01 -> DEPTH-02.
- Integration: KIT-01 -> genuine artifact exchange -> SVC-01 + VIS-01 -> prospective replay -> reproducible report.
- Evaluation: frozen scoring protocol -> hidden cases/baselines -> frozen fit/forecast -> score -> model update eligibility.

These chains intersect but should not unnecessarily serialize one another. In particular, missing phone depth does not block synthetic recovery; incomplete inverse fitting does not block capture or actual forward-geometry rendering. A blocked ticket remains explicitly unfinished.

5. Working integration gates

B integrates small reviewed commits into one designated integration branch. Follow repository deployment procedures when a runnable application change is in scope; a planning update alone does not merge or deploy the app. Keep feature worktrees separate and preserve the last passing integration state.

Gate	Required runnable evidence	Acceptance owner
G0: contract skeleton	Shared schemas and validation commands agree; owners and scoring protocol recorded.	B, with A scientific signoff
G1: real artifact exchange	Engine audio/geometry pair and actual synchronized capture validate and replay; provenance distinguished.	A engine acceptance; B capture acceptance

Gate	Required runnable evidence	Acceptance owner
G2: synthetic inference	Hidden-anatomy fit, fixed-anatomy baseline, candidate geometry display and independent score reproduce.	A numerical review; B independent evaluation
G3: human multimodal fit	Actual bundle enters FUSE-01; supported observations affect likelihood; missing depth and model mismatch reported.	A and B
G4: prospective loop	Supported prediction frozen before capture; outcome scored before update; stale/cancelled work excluded.	B; A forecast review
G5: evidence reproduced	Both humans replay benchmark and recorded run; equal-budget modality results, failures and limitations documented.	A and B

G2 may pass before the capture part of G1. The synthetic G4 path may be exercised before G3, clearly labeled synthetic. No synthetic result satisfies a human-data gate. Gates are evidence labels, not permission to misrepresent a partially integrated system.

Each handoff includes a commit, exact command, contract version, expected artifacts, actual checks and known failures. Integration checks should exercise real module boundaries, including error paths. B merges in dependency order; A reviews physical meanings and numerical claims before scientific changes enter the passing integration state.

6. Use extra agents for independent scientific challenges

Review and investigation agents do not share ownership of production files. Assign a specific hypothesis and artifact. Their independence helps expose errors; agreement between agents is not empirical validation.

Investigation	Required result	Lead
Identifiability	Distinct plausible anatomies with similar fit; candidate intervention predicted to distinguish them.	A
Anatomy/articulation separation	Stability across vowels with articulation varying; identify compensating parameters.	A
Capture compensation	Controlled microphone/room perturbations; determine whether inferred anatomy changes spuriously.	B, A reviews
Incremental depth value	Audio, audio+RGB, audio+RGB+depth comparison at equal evidence/compute budgets; report null results.	B
Model mismatch	Generate cases with altered physics/capture assumptions; test recovery and uncertainty failure.	A

Investigation	Required result	Lead
Evidence integrity	Attempt target leakage, stale-model updates, post-outcome forecasts and failed-trial exclusion.	B
Device reliability	Repeat capture/stop/permission-loss/replay on both available phones; preserve metadata.	B

For unresolved algorithms, permit bounded competing implementations behind the same interface in isolated branches. Predeclare a compute budget, development cases and selection metric. Tune only on development data; after selecting, evaluate once on fresh held-out cases to avoid selecting on the final test set. Promote one supported implementation and record why. Keep other branches as research evidence rather than merging all alternatives.

Schedule simulator workers, phones and benchmark compute explicitly. Pin versions, seeds, parallelism and budgets. If the native engine is stateful or not demonstrated thread-safe, isolate workers in processes and test deterministic reset behavior. More agents must not create uncontrolled compute contention or irreproducible comparisons.

7. Task board, dispatch and handoff contract

Use one shared repository-visible task board with IDs, owner, prerequisites, starting commit, contract version, acceptance experiment, branch/PR and status. Statuses: proposed -> ready -> running -> review -> integrated; blocked records the missing input and responsible owner. Only B marks repository integration complete after checks; scientific acceptance also requires A's signoff.

A task is ready when its input exists or its independent portion is explicitly bounded, one writer owns its files, and success can be checked. Split research/design and executable acceptance when only the former is unblocked. Coordinators revisit blocked dependencies at each handoff and dispatch the next ready task. Finishing agents take ready review or investigation tasks; they do not invent new architecture.

```
Task ID / objective:
Human lead / independent reviewer:
Owned files and explicit non-goals:
Starting commit / branch / worktree:
Input artifacts and contract version:
Prerequisites / what can proceed independently:
Acceptance experiment and required checks:
Compute or device budget:
Handoff commit / exact command / expected output:
Actual results / limitations / blockers:
Next consumer and dependency unblocked:
```

Agent starter instruction:

> Read revision 6, the lane assignment and shared contracts. You are not alone in this repository. Modify only your assigned files in your own worktree; do not revert others' changes or edit shared interfaces without their owner. Implement the bounded deliverable with real producers and explicit failure behavior. Separate research hypotheses, synthetic evidence and human observations. Run the acceptance experiment and report its actual result, including failure. Return a small commit, runnable commands, artifact provenance, contract version and remaining blockers. Do not expand scope, fabricate fitted anatomy or merge your own work.

8. Human checklists

Person A: scientific model lead

- [ ] Agree the first physiological hypothesis, supported parameter subset and scoring protocol.
- [ ] Dispatch PHY-01 and scientific feasibility tasks; obtain genuine forward artifacts.
- [ ] Review AUD-01 features and GEO-01 physical constraints with their owners.
- [ ] Dispatch INV-01, FUSE-01 and SVC-01 as prerequisites pass.
- [ ] Review anatomy/articulation separation, parameter recovery, candidate diversity and baselines.
- [ ] Dispatch PRED-01 and identifiability/model-mismatch investigations.
- [ ] Sign off numerical changes entering the integration queue.
- [ ] Review DEPTH-01/02 and publish supported conclusions and unresolved failures.
- [ ] Dispatch MOT-01 and integrate MOT-02; test control limits separately from anatomical limits.

Person B: experimental system and integration lead

- [ ] Establish KIT-01, file ownership, task board and integration queue.
- [ ] Dispatch native capture, acoustic measurement and independent evaluation tasks.
- [ ] Arrange device access and obtain CAP-01/02 timing/depth evidence.
- [ ] Dispatch service integration, actual geometry visualization and orchestration when ready.
- [ ] Keep hidden scoring truth separate from fitting/planning inputs; review leakage controls.
- [ ] Integrate reviewed commits and maintain the runnable cross-service path.
- [ ] Execute prospective and modality-comparison experiments with all attempts retained.
- [ ] Verify both humans can reproduce results and explain the evidence levels.
- [ ] Dispatch CUE-01 and LEARN-01; obtain cue review and demonstrate G6/G7 with actual evidence.

Both humans resolve cross-lane decisions in a short repository decision record, with affected contracts/tasks and migration owner. Preserve physiological inference, real evidence and independent scoring when reducing scope; reduce conversation polish, accounts, large-scale scraping and elaborate visual effects first. Anatomical recovery remains a hypothesis to test, not a completion claim inferred from agent output.

Embodied learning and dynamic physiology

Revision 6 addendum. The teacher must translate intended physical/acoustic changes into familiar actions, help the singer recognize their own sensations, and test whether they can reproduce and transfer the result. Initial mapping must capture movement and achievable control as well as static anatomy. These are integrated research and teaching capabilities, not a replacement for physiological inference.

1. A personalized action-to-physiology model

Maintain separate, linked estimates for shared anatomy, time-varying articulation, cue-conditioned motor control, acoustic response and reported sensation. Learn how this singer executes a familiar action in a specific pitch/vowel/level/posture context. A cue is a probabilistic input, not a command that sets a simulator parameter exactly.

The model should distinguish:

- Stable geometry and hypothesized physical limits.
- The comfortable movement envelope actually observed in a task.
- The subset the singer can intentionally reach and reproduce with a particular cue.
- Coordination, transition timing, repeatability and changes with practice.
- Sensor visibility, measurement uncertainty and competing physical explanations.

An unsuccessful attempt may reflect an unfamiliar cue, coordination, occlusion or sensor failure. It does not establish an anatomical restriction. An elicited gesture may succeed before voluntary recall succeeds. Track those separately. The maximum observed displacement is not the person's true physiological maximum, and new skill need not imply changing anatomical dimensions.

2. Familiar actions as motor-learning cues

Use an explicit loop: goal -> supported physical hypothesis -> familiar cue -> synchronized observation plus sensation report -> comparison with prediction -> feedback -> repeat -> fade cue -> transfer into singing. Explain the purpose of the experiment; do not deceive the singer about what the model knows.

Candidate cue family	Intended experience or experiment	Interpretation and implementation rule
Gentle breath with hands over comfortable side-rib/back regions	Notice expansion and release; connect perceived movement with a sustained easy sound.	Self-touch and external motion are observations, not confirmation of particular oblique activation, diaphragm position or correct support. Avoid maximal inhalation instructions.
Cough as a familiar abdominal sensation	User-proposed reference for locating an experienced abdominal response.	Retain as a specialist-review candidate, not an automated default exercise or repeated drill. It does not establish the muscle coordination desired in singing.
Duck-like quack or a small bright sound	Explore a reproducible twang-like acoustic contrast.	Candidate mnemonic, not proof of a unique narrowing or nasal configuration. Match pitch/vowel/level where feasible and measure the response.
Gentle cry-like or plaintive imitation	Explore an accessible quality/coordination contrast.	Do not label success as measured laryngeal or thyroid-cartilage tilt. Hidden movement remains a hypothesis unless independently measured.
Comfortable "ah", a yawn-like opening, tongue extension and return	Expose visible surfaces and sample movement trajectories.	Observe comfortable attempts, not forced maximum stretch. Yawning changes multiple structures; do not assume isolation of the palate.

These examples form a reviewable research library, not a prescription. Clinical voice guidance describes yawn-sigh as a facilitating technique; this does not validate every proposed mnemonic or its anatomical explanation. Twang research reports acoustic/source differences but does not establish that a duck imitation uniquely identifies a hidden anatomical action. Sources and evidence levels belong to each cue. ASHA voice guidance, Sundberg and Thalén, What is Twang?.

A singing teacher/voice specialist reviews the initial library and permissible variants. Keep coughing review-only, use comfortable attempts, and stop locally on pain, dizziness, strain or worsening voice. Do not encourage throat manipulation or put equipment inside the mouth. Vocal care guidance supports avoiding excessive vocal demands; this application does not diagnose respiratory or voice disorders. NIDCD voice care.

Astra selects and explains reviewed cues, adapts wording within their allowed variants, and asks what the learner felt. If the attempt fails, consider another cue, easier context or rest; do not merely repeat a technical anatomical instruction or demand more effort. New physical maneuvers need library review before runtime use.

3. Dynamic mapping with synchronized capture

Use repeated neutral -> comfortable gesture -> return trajectories. Begin with ordinary vowel changes, lip/jaw movement and comfortable tongue exposure. Add reviewed yawn-like and other contrasts where useful. Preserve task start/end, cue delivery, actual onset, returns, unsuccessful attempts, synchronization error and per-frame visibility.

CAP-01/02 records audio/RGB/depth throughout each trial. GEO-01/MOT-01 registers rigid head pose separately from articulation and estimates only supported visible trajectories. Do not rigidly fuse different tongue or palate poses into one static surface. The mouth can expose different surfaces across gestures, but hidden surfaces are not measured merely because a filled mesh looks complete.

For visible palate motion, track a repeatable visible region in a declared coordinate frame with confidence/occlusion flags. If a region cannot be identified consistently or depth is missing, do not output a millimeter range. Audio may constrain competing velopharyngeal-motion hypotheses, but does not make the inferred path a direct scan. Report observed excursion, repeat variability, cue dependence and confidence separately from hypothesized anatomical limits.

Breathing observations use a separate explicitly consented external torso view if needed; a face/mouth scan does not measure rib or abdominal motion. Separate views retain their own poses and trial identities. Self-reported back/side sensations do not require a torso video and remain useful as subjective feedback. The initial engine may not model respiratory mechanics: retain these observations and unsupported status without inventing muscle parameters.

4. Shared contracts and learning loop

Extend KIT-01 with the following versioned records. B owns schemas; A reviews physical semantics.

- CueDefinition: ID/version, wording and allowed variants, familiar action, intended hypothesis, supported engine variables, expected observables, evidence/source, review status, context, effort/rest/stop rules and alternatives.
- CueAttempt: cue/version, delivered wording, goal, model/prediction IDs, context, timestamps, repetition, capture references and execution deviations.
- SensationReport: learner's words and optional body-region selection, ease/effort, discomfort, confidence and whether the feeling is recognizable; explicitly subjective provenance.
- MotionObservation: trajectories, coordinate frame, visibility/quality, cue alignment, observed envelope and measurement uncertainty.

- ControlProfile: cue-conditioned reachable states, repeatability, transition timing, context dependence, learning history and competing explanations; distinct from anatomical parameters.
- TransferEvaluation: repeat without the full mnemonic, then use in a vowel/phrase or later session; acoustic error, supported movement agreement, effort and retention results.

A's motor model learns a distribution over executed articulation given cue, context and control history. PRED-01 marginalizes over that distribution for prospective forecasts. Post-capture measured movement can support a separately labeled conditional acoustic prediction; never rewrite a prospective forecast with outcomes. Update control, anatomy and subjective memory through separate evidence paths. A sensation report must not directly overwrite inferred anatomy.

B's teacher interface shows one actionable cue, an optional demonstration, what to notice, the learner's own remembered sensation, and predicted/observed differences. Display measured movement distinctly from hypothesized hidden motion. Let learners compare successful repetitions and save a personal mnemonic. Fade assistance and test transfer; immediate imitation alone is not learned control.

5. Ownership, dependencies and acceptance

Ticket	Lead / exclusive owner	Acceptance evidence
CUE-01 reviewed cue library and teacher flow	B; experiment/cues/, apps/web/coach/, tests/cues/	Versioned reviewed examples, alternatives and stop behavior; a learner can report sensation, replay an attempt and retrieve their own mnemonic.
MOT-01 dynamic control model	A; science/control/, tests/science/control/	Known synthetic trajectories/control variation recover within declared limits; distinguish uncertainty, occlusion and unsuccessful execution from anatomy restriction.
MOT-02 dynamic measured capture/fusion	B capture; A geometry/inverse via existing owners	Repeated real gesture/return trial replays with alignment and visibility; actual visible trajectory affects fitting; hidden/missing geometry stays labeled.
LEARN-01 recall, transfer and retention evaluation	B independent evaluator; evaluation/learning/	Compare cue variants against a reviewed fixed-cue baseline; freeze scoring/context; include failed trials and cue-free recall plus phrase transfer. Later-session retention is a separate result.

CUE-01 library/contracts and synthetic MOT-01 design can start alongside foundation work. Executable MOT-01 depends on PHY-01 and trajectory contracts. MOT-02 depends on CAP-02, GEO-01 and the inverse interface. Teacher flow begins with genuine artifacts; integrated cue selection depends on PRED-01 and ACT-01. LEARN-01 protocol can be prepared early, but its claims require actual learner evidence.

Add acceptance gates G6 dynamic mapping and G7 embodied transfer. G6 demonstrates repeated observed motion, contextual control estimates and explicit missing-data uncertainty. G7 demonstrates one goal -> hypothesis -> reviewed cue -> measured attempt -> sensation -> repeat -> cue-free/phrase transfer cycle, scored independently. An unsuccessful trial still produces a valid research record, but does not pass a learning-improvement claim. Compare cue-aware versus cue-agnostic execution forecasts and learned versus fixed cue selection; physiological accuracy still needs independent anatomical evidence.

Implement these prototype loops now, alongside reconstruction. Population coaching efficacy and detailed muscle/tissue explanations remain later validation work. The unique aim is to learn both the person's physical sound-producing system and how they can intentionally control it, then connect that understanding to sensations and skills they can use.

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